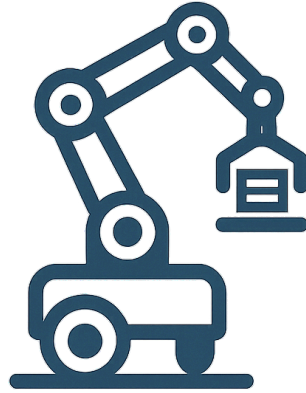




CATALYST

MRSD 2026 Team D: Catalyst Mobile Robot for Lab Automation

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2. Project Description

Modern research laboratories, especially in pharmaceutical, chemical, and biological domains, have widely adopted automation through robotic liquid handlers, spectrometers, and other specialized instruments to improve efficiency and reproducibility [1]. Yet a major bottleneck remains: material transfer between machines is still performed manually. Scientists must physically move wellplates, vials, and reagents between instruments, often many times per day, introducing opportunities for contamination, human error, and delays. Studies in clinical laboratory science show that up to 70% of laboratory errors originate in the pre-analytical phase, which includes manual handling, transport, and labeling [2]. Although reported in clinical settings, the same challenges apply across research labs that depend on human-mediated sample transfers.

This reliance on personnel prevents labs from achieving true end-to-end automation. Even highly automated facilities cannot run continuous 24/7 workflows if a human must be available to move materials between machines. Industries such as drug discovery, synthetic biology, and materials development require large scale experimental workflows; As labs scale up experiment volume and adopt more complex workflows, this mismatch between automated instruments and manual transfers becomes increasingly restrictive.

Cloud laboratories highlight this limitation clearly. These next-generation facilities allow scientists to design experiments remotely, with instruments operating autonomously under centralized software control. However, without automated sample transport, cloud labs still depend on technicians for routine handoffs, constraining throughput and increasing operational cost.

Catalyst addresses this automation gap by providing a mobile manipulator capable of autonomously transporting wellplates between instruments throughout a laboratory environment. By integrating navigation, perception, and manipulation into a single platform, Catalyst enables fully automated multi-instrument workflows without human intervention. It performs transfers with consistency and repeatability, reducing contamination risks and handling errors that come with manual transport. Because the system can operate continuously, it supports true around-the-clock experimentation, increasing throughput and accelerating scientific progress. Catalyst is also designed to integrate into existing laboratory spaces without requiring major infrastructure changes, making the system scalable and compatible with standard lab environments.

Catalyst transports standard wellplates, the most common container in modern wet labs, ensuring compatibility with liquid handlers, incubators, readers, and sensing instruments already in widespread use. The system does not interpret or analyze experimental output; instead, it integrates with cloud-lab backends and leaves scientific evaluation to specialized instruments.

By enabling safe, precise, and autonomous movement of samples, Catalyst removes one of the last remaining barriers to fully automated scientific workflows.

3. Use Case

Inside a modern cloud laboratory, an automated, remotely operated workspace where instruments can run experiments without human presence, researchers interact with the environment entirely through software. In this setting, every machine is networked, capable of reporting its status, and able to signal when a task is complete. The cloud lab also automatically logs all experimental outputs for the user. Our system is deployed within this environment, and relies on these capabilities: instruments inform the robot when they are done, and the cloud lab backend handles all experimental results.



Figure 3.1: Catalyst in the Bakery Square Cloud Lab

A user begins by submitting a workflow that specifies the sequence of operations required for their experiment. In this example, the workflow involves two primary instruments, a liquid handler and a spectrometer. Once the workflow is received, our system interprets the steps and begins coordinating the robot's movement through the lab.

The robot navigates through the workspace to the storage area, pictured in Figure 3.1, identifies the required vial or well plate, and securely grasps it. It then travels to the liquid handler, where it loads the sample for automated pipetting or reagent mixing. While the liquid handler executes its programmed routine, the robot waits at a safe distance. When the task is finished, the cloud lab signals completion, allowing the robot to continue to the next step.

After retrieving the sample from the liquid handler, the robot proceeds toward the spectrometer, the final sensing instrument in this workflow and the machine responsible for producing experimental results. The robot positions the sample in the spectrometer's loading area, after which the instrument performs the required measurements. Throughout this process, our system does not read, interpret, or analyze any sensed data. Instead, the spectrometer sends all measurement results directly to the cloud lab backend, ensuring the user receives a complete log of the experiment.

Once the sensing step is complete, the workflow concludes. Because every experiment in our system is designed to end at a sensing machine, the cloud lab provides the final output, enabling the robot to focus exclusively on safe, accurate, and reliable transportation between instruments

4. System-Level Requirements

We have established several requirements that support the system’s central goal of usefully transporting wellplates around a wet lab.

The requirements for this system are categorized as mandatory or desirable, and performance or non-functional, as shown in table 4.1. The coding system reflects this. For example, a code of “M.P.x” refers to a mandatory performance requirement.

Table 4.1: Requirements for Catalyst System

Code	Description	Performance Metric (if applicable)
M.P.1	Localize wellplate	± 2 cm of nominal location
M.P.2	Grasp wellplate	90% reliability
M.P.3	Pick wellplate from instruments	1 min pick cycle
M.P.4	Place wellplate in instruments	1 min place cycle
M.P.5	Avoids obstacles	100% stopping rate for obstacles
M.P.6	Localize itself within lab space	± 10 cm of nominal location
M.P.7	Navigate between instruments	20 cm/s transit speed
D.P.1	Communicate experiment results	5 min update frequency
M.N.1	Safe operation around lab personnel	
M.N.2	Does not spill liquids	
M.N.3	Fits between lab aisles	
D.N.1	Easily recharged	

4.1 Wellplate Interaction Requirements

The wellplate is a reagent container that is common to many experimental workflows in biology and chemistry labs, so we have several requirements pertaining to interacting with them. Detecting wellplate locations, grasping them securely, and successfully replicating how a human would insert and remove them from a lab instrument are all behaviors that our system requires to be useful to our target demographic of experimentalists.

4.2 Mobility Requirements

To ensure the robot can productively move wellplates around the laboratory, it also needs to avoid obstacles, localize itself among the instruments, and successfully navigate between locations of interest.

4.3 Nonfunctional Requirements

To complete our system's compatibility with the laboratory environment, the non-functional requirements ensure that the system will integrate effectively with the work experimentalists do. Safe operation, space efficiency, and protecting the reagents for experimental integrity are all required for the system to be trusted with rigorous scientific experimentation.

5. Functional Architecture

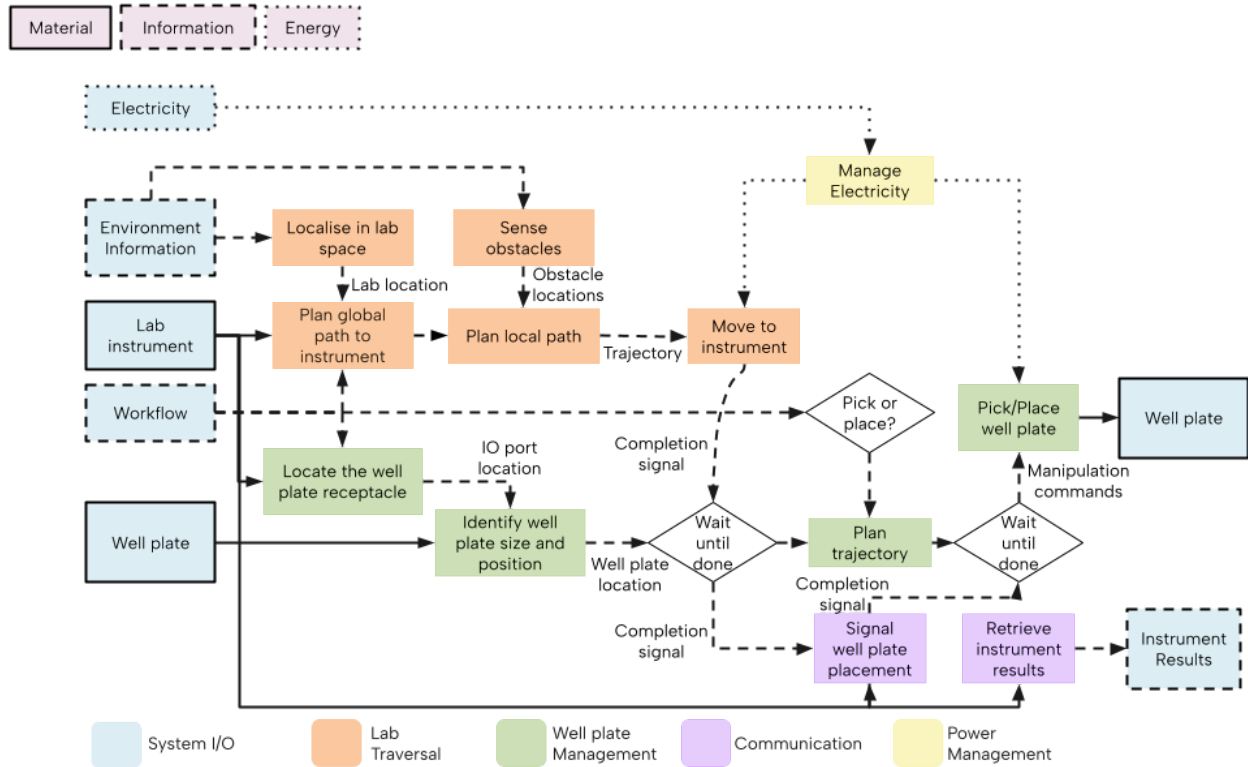


Figure 5.1: Functional Architecture

The functional requirements are fully represented in the functional architecture, with some being split into a sequence. Per the legend in the top left of figure 5.1, outputs are classified as material, energy, or information by the type of line stroke. Functions are grouped into four implementation-generic subsystems: Lab Traversal, Well Plate Management, Communication, and Power Management. These subsystems have few, but critical interconnections since they independently fulfill different functional requirements. This structure supports parallel development and execution of subsystems. Table 5.1 demonstrates which functional requirements each subsystem fulfills:

Table 5.1: Functional Requirements by Subsystem

Lab Traversal	Well Plate Management	Communication
M.P.5: Avoids obstacles M.P.6: Localize itself within lab space M.P.7: Navigate between instruments	M.P.1: Localize wellplate M.P.2: Grasp wellplate M.P.3: Pick wellplate from instruments M.P.4: Place wellplate in instruments	M.P.3: Pick wellplate from instruments M.P.4: Place wellplate in instruments D.P.1: Communicate experiment results

6. Trade Studies

6.1. System-Level Trade Study

					
Spec/Criteria	Weight	Human-based	Fixed Workcell	Mobile Manipulator	Gantry Automation
Works Autonomously	25%	1	3	5	4
Flexible Across Instruments	20%	5	2	5	2
Scalability & Repeatability	25%	2	4	4	5
Cost	10%	3	3	3	2
Speed of Operation	20%	3	4	4	5
Weighted Average	100%	2.65	3.25	4.35	3.85

Table 6.1 System Level Trade Study

The two main criteria for our system level trade study are:

1. **Works Autonomously:** The purpose of lab automation is to reduce manual intervention in routine, high-volume experiments. True autonomy ensures:
 - **End-to-end workflows:** The robot has to read a scientist-defined protocol (e.g., “Incubator → Reader → Washer → Freezer”), schedule steps, and execute them without human “bridging” between machines.
 - **Machine interaction:** Autonomy means more than navigation – it includes opening/closing doors, placing vials/plates in receptacles, pressing buttons/GUI substitutes, and handling barcodes/IDs.
 - **Unattended operation:** Experiments often require overnight sequences or precise timing between machines. Without real autonomy, a human has to “babysit” hand-offs, which defeats the point of automation.
2. **Scalability & Repeatability:** Modern wet labs demand high-throughput, consistency, and reproducible outputs across many samples and instruments. A scalable, repeatable system offers:
 - **Running many workflows in parallel:** More samples, more plates, or more experiments per day with the same staff.
 - **Consistent inter-device hand-offs:** Repeatedly placing vials in the exact receptacle position, correct orientation, and within timing tolerances so downstream machines behave reliably.
 - **Adding/Changing instruments & workflows:** As the lab adds a new analyzer or modifies the workflow, the system should adapt without a full re-build.

In the context of vial and well-plate transfer across multiple laboratory instruments, the human-based system scores lowest for autonomy (1/5) because every hand-off between machines depends on an operator’s availability and judgment, preventing unattended workflows and strict timing control, and it scores poorly on scalability and repeatability (2/5) since human technique, fatigue, and person-to-person variability limit both consistency and high-throughput execution. A fixed workcell performs well inside its enclosed reach zone, achieving moderate autonomy (3/5) by automating transfers only among machines physically clustered around the arm, but still requiring humans to bridge any devices outside the cell; however, it delivers strong repeatability (4/5) because arm trajectories and receptacle placements are precise once programmed, though

scaling demands new cells or reconfiguration. A mobile manipulator achieves full autonomy (5/5) because it can navigate across the lab, dock with multiple instruments, and execute complete experiment workflows without human transport of vials, enabling truly unattended multi-device operation; its scalability and repeatability are also high (4/5) since it adapts to new instruments via software mapping rather than infrastructure changes, though its performance is slightly constrained by navigation error and perceptual uncertainty compared to fully constrained systems. Finally, gantry automation provides high autonomy (4/5) for any machines located under its rails, executing reliable transfers at scale, but it requires physical infrastructure modification to extend beyond its footprint, making it less flexible than a mobile system; it earns the highest repeatability score (5/5) because its constrained, rail-guided motion yields near-perfect and highly stable placement accuracy, ideal for large labs with fixed layouts and massive sample throughput.

6.2. Subsystem-Level Trade Studies

To further support the system-level decision of deploying a mobile manipulator for vial and well-plate transfer across laboratory instruments, we conducted two subsystem-level trade studies focusing on the enabling technologies that most directly impact reliable autonomy. The first study evaluated localization methods, which determine how the robot navigates within the lab and accurately docks to individual machines, comparing four approaches: magnetic tape guidance, fiducial marker-based localization, LiDAR + IMU fusion, and vision-based odometry. The second study examined end-effector/gripper mechanisms, since secure vial handling and precise placement on instrument nests are critical for workflow repeatability. Four gripping solutions were analyzed: parallel jaw grippers, vacuum-based grippers, adaptive multi-finger grippers, and tool-changing systems designed for multi-end-effector use. Together, these subsystem evaluations allow us to identify the most robust navigation and manipulation technologies needed to deliver reliable, autonomous operation in a multi-instrument laboratory environment.

6.2.1 Subsystem Level Trade Study - Localisation Methods

					
Spec/Criteria	Weight	Magnetic Tape	Fiducial Marker	LIDAR + IMU	Visual Odometry
Localisation Accuracy	20%	5	5	4	5
Flexibility to Map Changes	20%	1	3	4	4
Maintenance Effort	15%	1	3	4	3
Sensor Cost	10%	5	4	1	3
Scalability	20%	1	3	5	4
Works on Obstructed Lab Spaces	15%	2	3	5	3
Weighted Average	100%	2.35	3.5	4.05	3.4

Table 6.2 Subsystem-Level Trade Study: Localisation Methods

For autonomous vial transport in a dynamic laboratory environment, localization accuracy and flexibility to map changes are critical because the robot must repeatedly dock to instrument interfaces with millimeter precision and adapt as equipment is moved, added, or reconfigured. Although magnetic tape guidance offers high positional accuracy (5/5) due to perfectly constrained paths, it scores very low in flexibility (1/5) since any workspace change

requires physically re-laying tape, making it unsuitable for evolving labs; it also suffers from poor scalability (1/5). Fiducial marker localization provides similarly high accuracy (5/5) and moderate flexibility (3/5) because markers can be added or relocated with minimal effort, though it introduces installation and maintenance overhead and is less robust in cluttered areas. LiDAR + IMU fusion offers the best balance overall, achieving strong accuracy (4/5) along with high adaptability (4/5) as it builds and updates maps without relying on fixed environmental features; it also scales well (5/5) and can navigate obstructed lab spaces reliably. Visual odometry, while capable of high accuracy (5/5) in well-lit and feature-rich environments, is more sensitive to lighting, occlusions, and surface textures, reducing robustness in cluttered lab spaces (3/5) and yielding moderate flexibility (4/5). Therefore, LiDAR + IMU emerges as the optimal localization approach, offering reliable precision and adaptability for multi-instrument laboratory workflows.

6.2.2 Subsystem-Level Trade Study - Grippers

					
Spec/Criteria	Weight	Adaptive Multi Finger	Vacuum	Parallel Jaw	Tool Changing System
Compatibility with Labware	25%	3	2	4	5
Customizable Fingers	20%	1	1	5	5
Ease of Integration	20%	3	2	5	1
Cost	10%	2	5	4	3
Positioning Precision	25%	3	2	4	5
Weighted Average	100%	2.5	2.1	4.4	4

Table 6.3 Subsystem-Level Trade Study: Gripper Strategy

For autonomous vial and plate transfer between laboratory instruments, compatibility with standardized labware and precise positioning during placement are essential, since the robot must reliably integrate with varied receptacle geometries without dropping, tilting, or misaligning wellplates at instrument interfaces. Parallel jaw grippers score highest overall because they are inherently well-suited to rigid, repeatable gripping of plate edges (4/5 for compatibility) and provide consistent alignment during placement (4/5 positioning precision), while also offering high customizability of fingers (5/5) and easy integration with commercial robot arms. Vacuum grippers, though inexpensive and easy to mount, score poorly in both compatibility (2/5) and precision (2/5) because wellplates don't have suitable areas to be gripped with a vacuum gripper, and adding a compressor is a major design addition. Adaptive multi-finger grippers accommodate irregular objects but add unnecessary capability for structured labware, leading to moderate compatibility (3/5) and positioning precision (3/5) while increasing mechanical complexity and cost. Tool-changing systems offer maximum compatibility and precision (5/5 each) by swapping between multiple specialized tools, but their lower ease of integration (1/5) and higher cost/maintenance overhead (3/5) make them better suited to very high-throughput facilities rather than a flexible research-scale environment. Consequently, the parallel jaw configuration provides the best balance of repeatability, customization, cost, and direct suitability for labware, making it the optimal choice for our mobile manipulator.

7. Cyberphysical Architecture

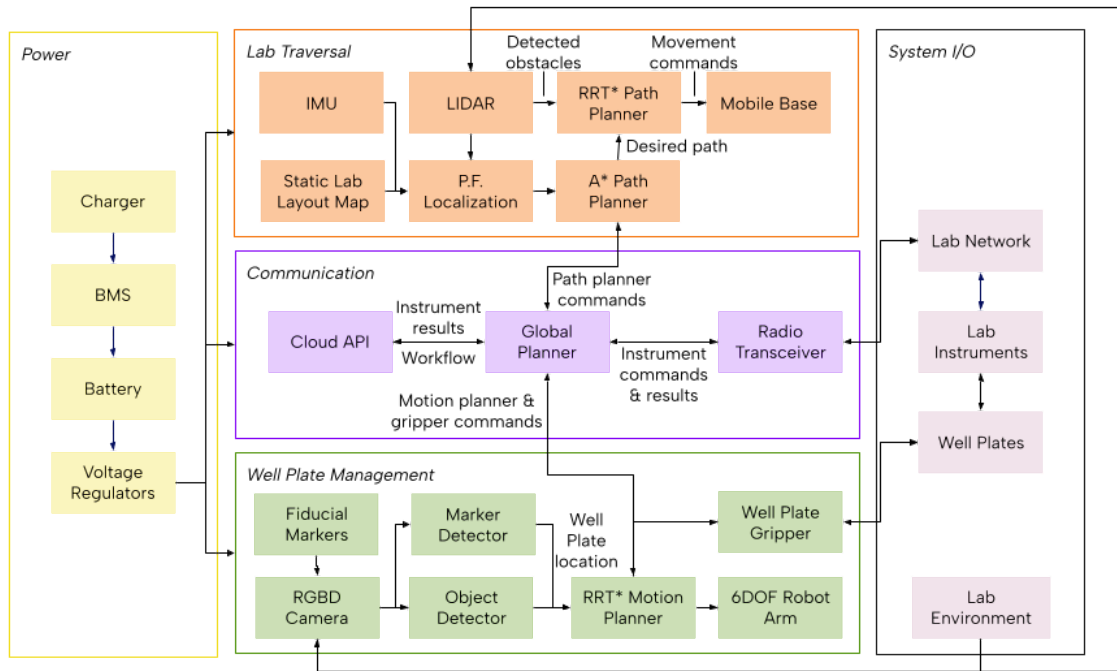


Figure 7.1: Cyberphysical Architecture

Table 7.1: Cyberphysical Choices by Subsystem

Lab Traversal (Mobile Base)		Well Plate Management (Manipulator)		Communication (Central Compute)	
Function	Components	Function	Components	Function	Components
Localize in lab space	IMU, LIDAR, P.F. Localization	Locate well plate IO port	Fiducial Markers, RGBD Camera, Marker Detector	Signal well plate placement	Global Planner, Network Antenna
Sense obstacles	LIDAR, (RGBD Camera, if necessary)	Identify well plate position	Fiducial Markers, RGBD Camera, Marker Detector	Retrieve instrument results	Global Planner, Network Antenna
Plan global path	A* Path Planner	Identify well plate size	Fiducial Markers, RGBD Camera, Marker Detector, Object Detector		
Plan local path	RRT* Path Planner	Plan trajectory	RRT* Motion Planner		
Move to instrument	Mobile Base	Pick/place well plate	6DOF Robot Arm, Well Plate Gripper		

The cyberphysical architecture depicted in Figure 7.1 was developed using the hardware/software choices from the trade studies and the relationships defined by the functional architecture to create a feasible implementation of a system to fulfill all requirements. Table 7.1 demonstrates how the functional architecture maps onto the cyberphysical architecture for the three subsystems with the most associated functions. The system combines the capabilities of several sensors to increase accuracy and robustness of critical functions such as localization and obstacle avoidance, making the system safer and more reliable.

The particle filter, A*, and RRT* algorithms are utilized for localization and planning due to their flexibility, despite their stochastic nature. They are all widely used within robotics, so software packages for these algorithms already exist, which reduces the risks of developing and integrating new software algorithms.

Going forward, subsystems will be referred to as the titles in parenthesis in Table 7.1, identified by the most important hardware component the subsystem contains, rather than by what type of functions the subsystem performs. The following section describes each subsystem in detail.

8. Subsystem Descriptions

To facilitate a smooth engineering development process, we have divided our system into four subsystems: manipulator, mobile base, central compute, and power management.

8.1 Manipulator Subsystem

To interact with wellplates (and potentially more lab supplies, in the future), our system will employ a custom end-effector and off-the-shelf robotic arm, collectively referred to as the manipulator subsystem. The hardware is paired with localization and planning capabilities to enable it to sense and interact with well plates, in the form of a camera to sense the positions of well plates, object detectors to localize the well plates, and motion planners to orchestrate physical manipulation.

8.2 Mobile Base Subsystem

The wellplates will be conveyed around the laboratory with the mobile base subsystem. The hardware we have selected for this subsystem is an electric wheelchair base that is powered by two 12V lead-acid batteries, a pair of brushed DC motors, and conventional optical encoders, all driven by a Roboteq motor driver. The mobile base will be augmented with LiDAR and IMU sensors to enable perception and localization for robotic decision making.

8.3 Central Compute

Centralized compute resources are shared between the two previous systems on an on-board computer, where the navigation, perception, and manipulation software stacks will run. In addition, there is a server which coordinates the interaction between the mobile manipulator and lab instruments, along with managing the workflows submitted by experimentalists. Because the lab is networked, there are no restrictions on where the server must be located, as long as it is network-connected. The compute subsystem will leverage the Robot Operating System (ROS) to coordinate concurrent tasks between other subsystems.

8.4 Power Management

Power will be shared between both the base and the manipulator. Power management will be a major design effort, as the current power infrastructure on the mobile base is not sufficient to also serve the arm and the main computer. The lead acid batteries in the mobile base will be integrated with the main power management system so the end user may charge all powered systems conveniently. The final power management subsystem will contain a single charging interface, a battery management system to ensure safe and efficient battery charging, batteries for both the base and the arm, and voltage regulators to ensure stable power delivery to the other subsystems.

9. Project Management

9.1 Work Plan and Tasks

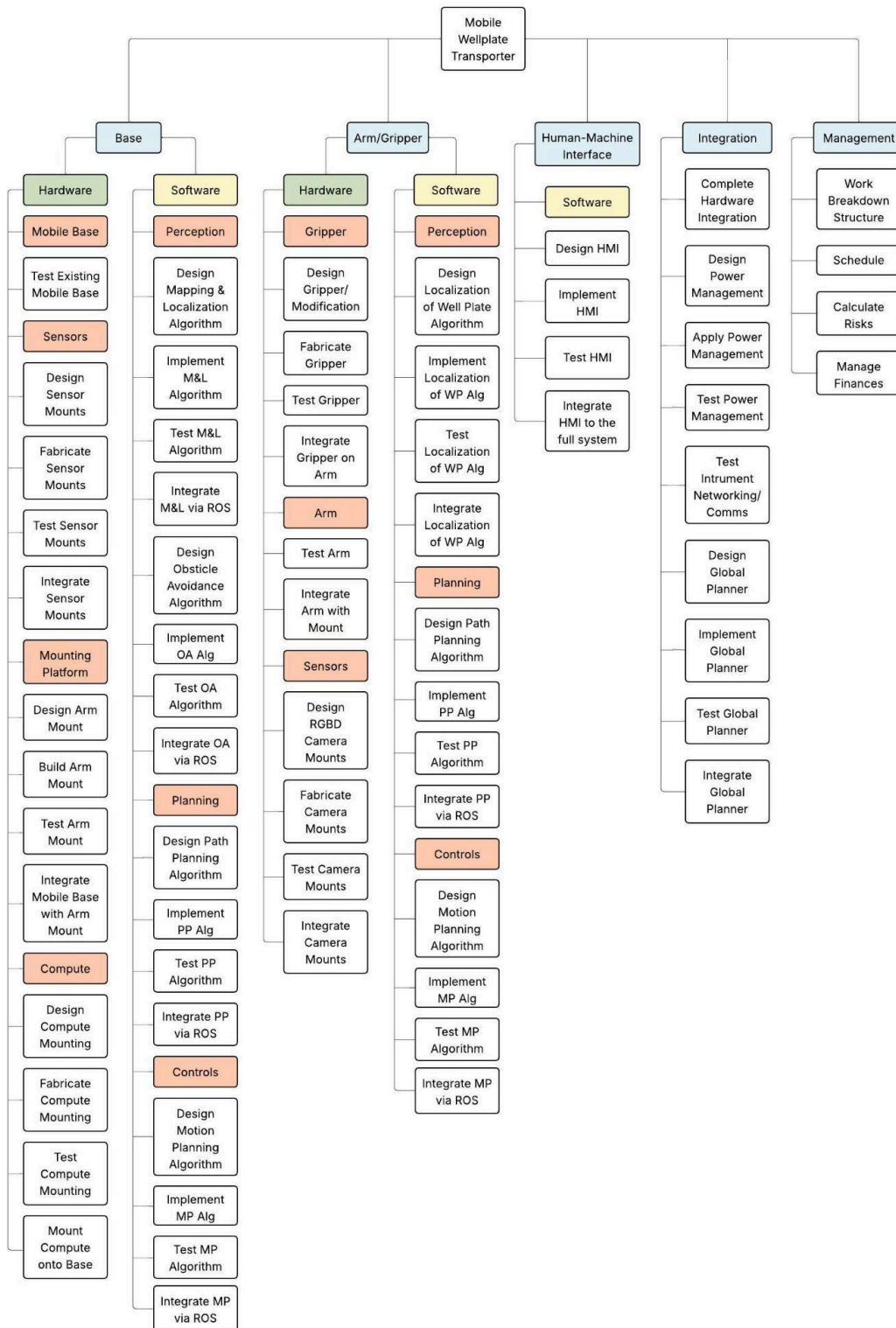


Figure 9.1: Work Breakdown Structure

Figure 9.1 outlines the Work Breakdown Structure (WBS) for Catalyst. The WBS divides the workload into manageable components with 5 major sections: Base, Arm/Gripper, Human-Machine Interface, Integration, and Management. This structured approach not only aids in organizing tasks but also facilitates the creation of an efficient work schedule.

Key milestones:

- **End of Fall Semester:** All major hardware components procured.
- **February 12:** Base and manipulator subsystem hardware created and validated.
- **February 26:** Base controls and arm controls functioning
- **February 20:** Gripper capable of picking up a well plate; base sensors (encoders and camera) successfully streaming data through ROS.
- **March 6:** Integrated testing of arm and base to ensure combined operation does not degrade subsystem performance.
- **March 13:** Manipulator achieves reliable well-plate pickup; base demonstrates autonomous navigation without trajectory planning.
- **April 23:** Full Spring Validation Demonstration.

9.1.1 Progress Report Goals

There are two progress reports mid-semester: 12 February and 26 February. Our goal for the first progress report is to have our hardware fabricated, barring final subsystem integration, and be ready to start software development. Our goals for the second progress report is to have control algorithms for the base and arm running, and to have successful object detection for the wellplates via the manipulator sensor.

9.2 Gantt Chart Schedule

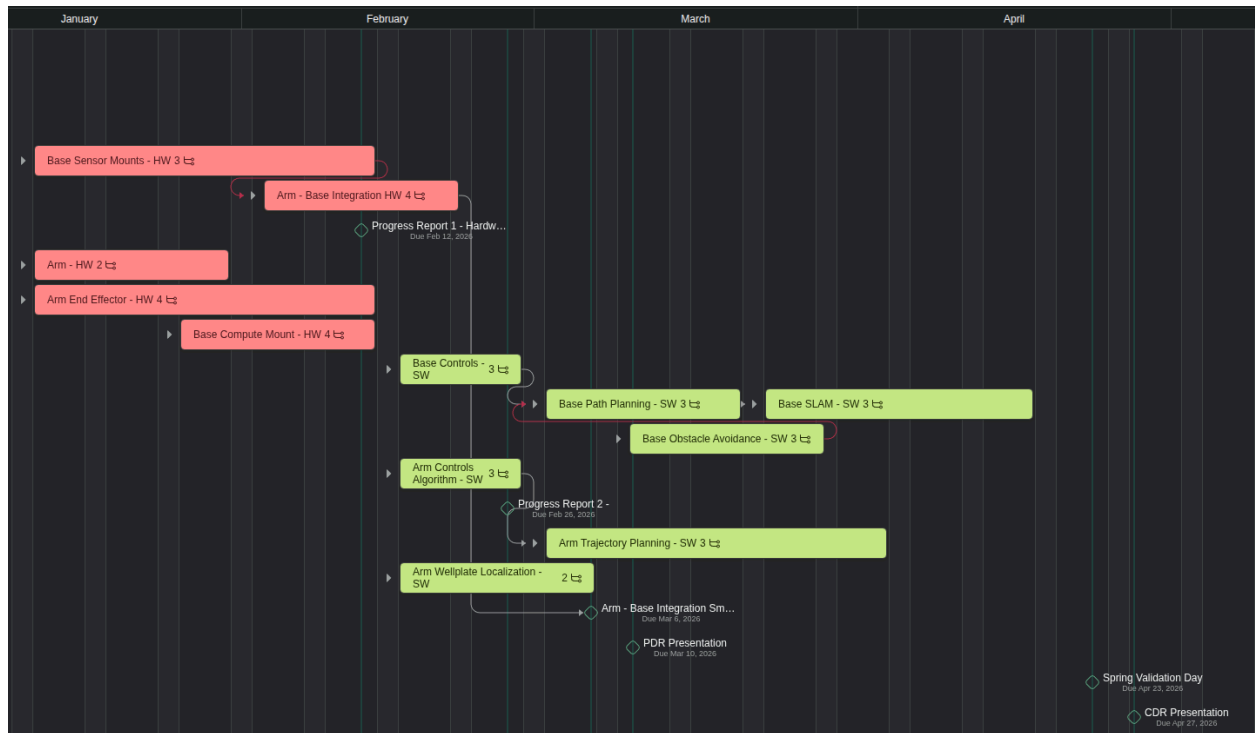


Figure 9.2: Spring Work Schedule

9.3 System Validation Demonstrations

The System Validation Demos are key points of the project in which we show project progress towards the ultimate goal of autonomous lab material transfer.

9.3.1 Spring Validation Demonstration

The aim of the Spring Validation Demo (SVD) is to show the completion of our two major subsystems working completely independently of each other. The base should be able to navigate to instruments and avoid obstacles and the manipulator should be able to pick and place well plates from at least 2 instruments with a 90% accuracy.

Location: Bakery Square Lab

Equipment: Catalyst Mobile Base, Catalyst Manipulator, TV for Presentation, Junior Liquid Handler, PHERAstar FSX Spectrometer

Procedure:

Table 9.3: Spring 2026 Validation Procedure

ID	Description	System Requirement(s) Met
SV.1	Introduction	N/A
SV.2	Base starts a specified distance from a Junior Liquid Handler and successfully navigates to the instrument with ± 10 cm accuracy.	M.P.7
SV.3	Base successfully navigates around humans in the way	M.P.5
SV.4	Stationary manipulator successfully picks and places wellplates from Junior Liquid Handler	M.P.2, M.P.3
SV.5	Stationary manipulator successfully picks and places wellplates from PHERAstar FSX Spectrometer	M.P.2, M.P.3

9.3.2 Fall Validation Demonstration

The aim of the Fall Validation Demo (Fall Validation Demo) is to demonstrate the full working system. We anticipate that for FVD, we have full hardware integration of the manipulator and base, full networking of the Catalyst robot and the lab instruments, and that we have an orchestrator that commands both the robot and instruments which a user can interact with using a Human Machine Interface (HMI).

To run the demo, the user will input the desired workflow into the HMI. The robot, starting from its given starting position, will traverse to the well plate station and stop. Then, it will pick up a well plate and travel to the first instrument. After stopping in front of the instrument it will place the well plate into the designated position and communicate with the server that it has done so. Then the instrument will perform whatever task it is meant for and communicate with the server that it has done so. Catalyst will then pick up the well plate from the designated location and repeat the traversal and manipulation workflow for the rest of the experiment. Once the last instrument has been run, the final results of the experiment will be sent to the HMI where the user can view the results of the experiment and the robot will place the completed well plate in a designated area.

- Location: Bakery Square Lab
- Equipment: Catalyst Mobile Manipulator, TV for Presentation, Junior Liquid Handler, PHERAstar FSX Spectrometer, possibly other lab instruments in Bakery Square Lab

9.4 Responsibilities of Members

Table 9.4: Team Member Responsibilities

Team Member	Primary Responsibility	Secondary Responsibility
Srishti Gupta	Mobile Base Subsystem	Obstacle Avoidance
Shua Halle	Mobile Base Subsystem	Localization
Andrew Noble	Mobile Base Subsystem	Sensor Fusion
Risha Ranjan	Manipulator Subsystem	Project Management, Finance Management
Aman Tambi	Manipulator Subsystem	Gripper Design

9.5 Provisional Parts List and Budget

Table 9.5: Initial Parts and Cost List

Part No.	Part Name	Cost Per Unit	Unit Quantity	Total Cost
1	Wheelchair Batteries	175	2	350
2	Aluminum extrusion rails (2020/2040)	25	2	50
3	Anti-slip rubber pads	10	1	10
4	Custom/commercial well-plate gripper fingers	40	1	40

5	Force/torque sensor or load cell	60	1	60
6	IMU module	20	1	20
7	Calibration checkerboard	10	1	10
8	Cables and Connectors	15	2	30

9.6 Risk Management

Table 9.6: Top Five Risks and Mitigations

Risk	Consequence	Likelihood	Mitigations
Difficult to gain access the Bakery Square Cloud Lab (BSCL)	2	3	1. Communicate with sponsor 2. Use mock setup
Manipulation reliability is harder than expected	2	5	1. Test gripper as early as possible 2. Ask subject matter experts, we have contacts at NREC who have accomplished this task
Integration is harder than expected	3	4	1. Integration test in the Spring so we aren't caught by surprise in the Fall 2. Change scope to stationary arm
Cannot network with instruments as expected	2	2	1. Communicate with NREC/ BSCL 2. Change scope to remove networking
Hardware gets damaged	3	3	1. Work in pairs to prevent mistakes 2. Purchase spare parts 3. Change scope

10. References

- [1] I. Holland and J. A. Davies, “Automation in the Life Science Research Laboratory,” *Frontiers in Bioengineering and Biotechnology*, vol. 8, article 571777, Nov. 2020. doi: 10.3389/fbioe.2020.571777.
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